Group 16

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1. Topic and Research Question

1.1 Background and Motivation

A well-known baseline in climate economics is that national wealth (GDP per capita) correlates positively with carbon dioxide (CO_2) emissions. However, asking solely whether GDP correlates with CO_2 yields an overly simplistic answer and masks dramatic divergences between countries at similar stages of economic development. Over the past three decades, several high-income nations have reported declining domestic CO_2 emissions alongside continued GDP growth; a phenomenon termed "Green Decoupling."

This project investigates the underlying mechanisms behind these national trajectories by exploring whether countries lowering their territorial emissions are achieving genuine decarbonization through renewable energy transitions, or whether they are shifting heavy manufacturing abroad and importing carbon-intensive goods from industrializing nations.

1.2 Primary Research Question

"How have nations managed to grow their economies while lowering domestic CO_2 emissions since 1990? Through genuine renewable energy transitions, or by shifting heavy industry and carbon footprints abroad?"

1.3 Sub-Questions for Visual Exploration

To communicate these findings effectively across an interactive dashboard the analysis is guided by three sub-questions:

- **Structural Transformation Over Time:** Which countries have experienced the greatest change in recent years regarding their carbon intensity (CO_2 per unit of GDP), and how closely does this trajectory align with renewable energy

adoption versus the shrinking of domestic manufacturing in favor of service sectors?

- **Production vs. Consumption Footprints (Carbon Outsourcing):** How does carbon consumption differ across countries when comparing domestic territorial emissions against trade-adjusted lifestyle emissions?
- **Trade and Demographic Correlates:** How do national trade profiles (exports and imports as a share of GDP) and energy mixes relate to country-specific parameters such as population density and real purchasing power?

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2. Datasets and Data Integration Strategy

2.1 Overview of Data Sources

While a standalone emissions dataset provides detailed greenhouse gas metrics, it lacks the macroeconomic, industrial, and demographic variables necessary to explain why emissions shift. Therefore, we construct a custom multidimensional dataset by merging two authoritative open-data repositories:

- **Our World in Data (OWID) CO_2 and Greenhouse Gas Emissions Dataset** (owid-co2-data.csv), which compiles historical emissions and energy data from the Global Carbon Project and the Energy Institute Statistical Review of World Energy.
- **The World Bank – World Development Indicators (WDI)** (World Bank DataBank), which provides standardized global socio-economic, industrial, trade, and demographic indicators.